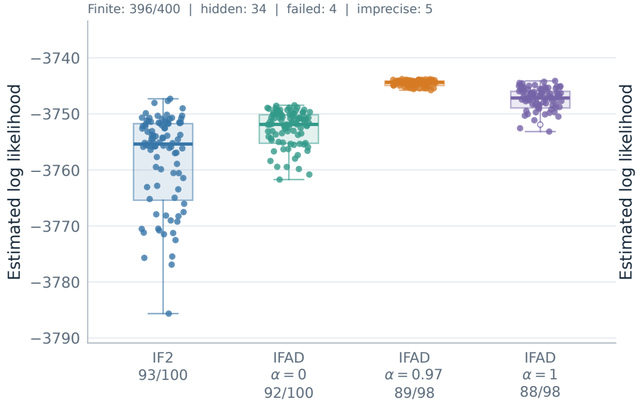
Dacca: log likelihood estimates

Outliers hidden using 1.5 IQR within each package-method group. Vertical scales differ between packages.

PyPOMP and Cuthbert: 100 initial values per method. Other packages: 6 initial values x 2 replicates. PyPOMP PF likelihood evaluation, J=5,000.

PyPOMP

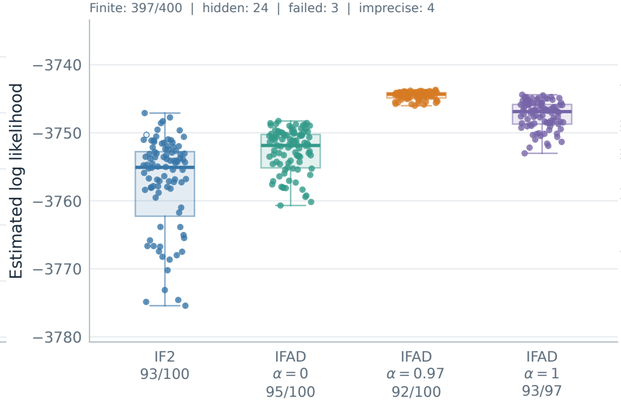
Finite: 396/400 | hidden: 34 | failed: 4 | imprecise: 5



IF2: functional.mif
IFAD: functional.mif then functional.train
Optimizer: pypomp.Adam

Cuthbert

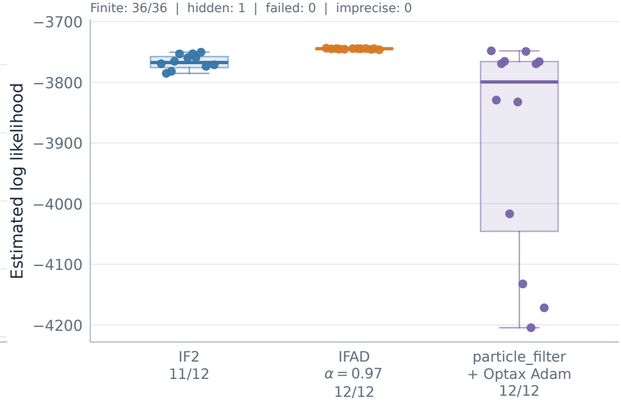
Finite: 397/400 | hidden: 24 | failed: 3 | imprecise: 4



PF: particle_filter.build_filter
Added here: IF2 / IFAD and gradient correction
Optimizer: pypomp.Adam

PFJAX

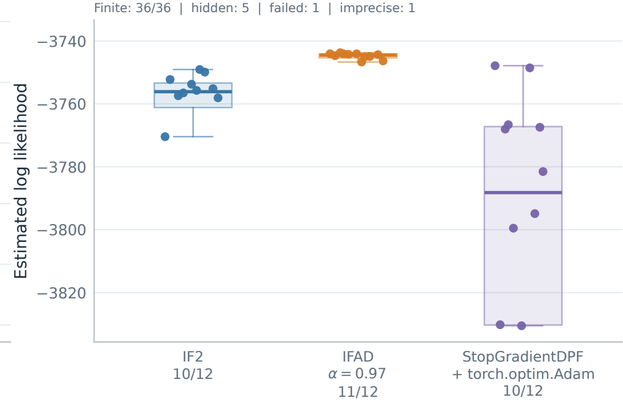
Finite: 36/36 | hidden: 1 | failed: 0 | imprecise: 0



PF: particle_filter; IF2: resample_multinomial
Added here: IF2 / IFAD and gradient correction
Optimizer: optax.scale_by_adam

PyDPF

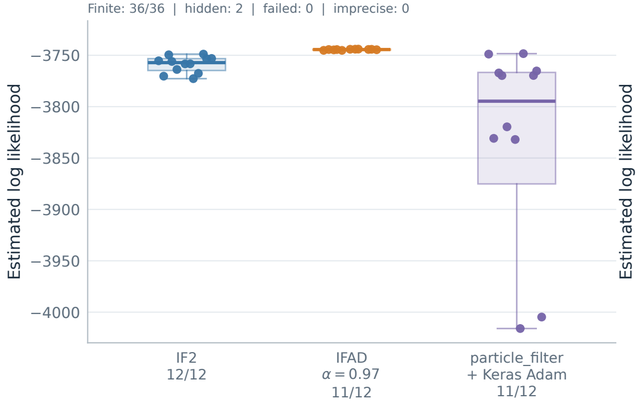
Finite: 36/36 | hidden: 5 | failed: 1 | imprecise: 1



Direct alpha=1: StopGradientDPF
IFAD: ParticleFilter + added alpha=.97 correction
IF2 added here; optimizer: torch.optim.Adam

TFP

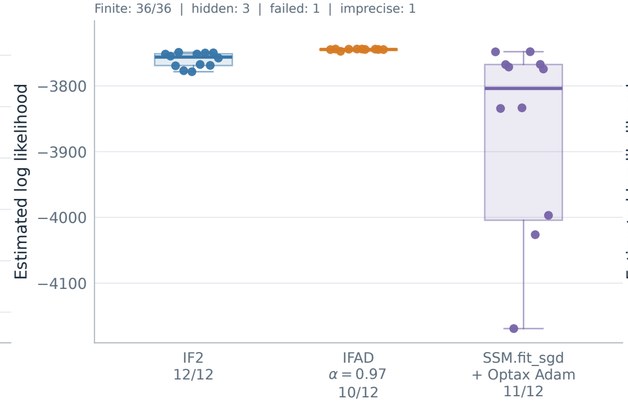
Finite: 36/36 | hidden: 2 | failed: 0 | imprecise: 0



PF: particle_filter(unbiased_gradients=True)
IF2: sequential_monte_carlo + added perturbations
Added alpha=.97 correction; tf.keras.optimizers.Adam

Dynamax

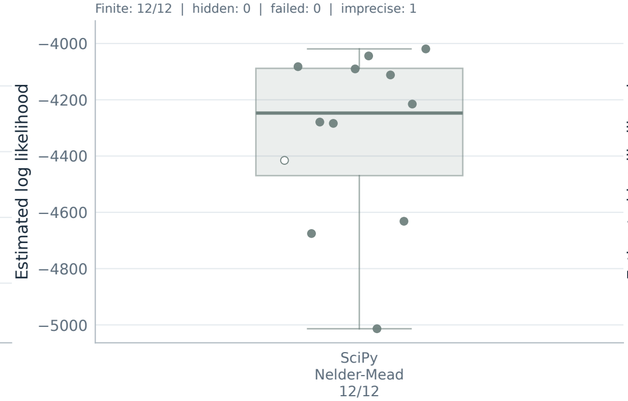
Finite: 36/36 | hidden: 3 | failed: 1 | imprecise: 1



Optimization interface: SSM.fit_sgd
Added here: Dacca PF / IF2 / IFAD
Optimizer: optax.scale_by_adam

particles

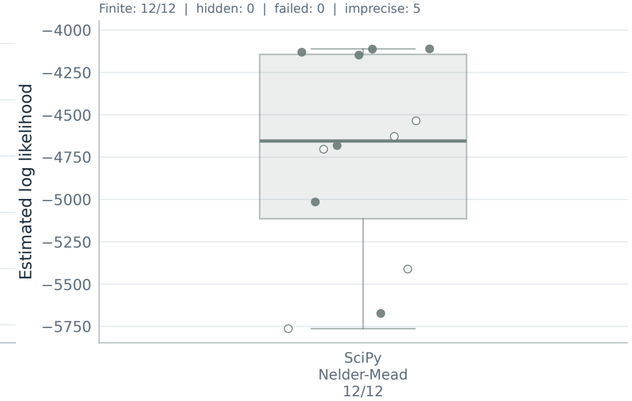
Finite: 12/12 | hidden: 0 | failed: 0 | imprecise: 1



PF: SMC, via a subclass that records likelihoods
Added likelihood objective; scipy.optimize.minimize (method="Nelder-Mead")

pypfilt

Finite: 12/12 | hidden: 0 | failed: 0 | imprecise: 5



PF: fit, used to run particle filtering
Added likelihood objective; scipy.optimize.minimize (method="Nelder-Mead")

● MCSE and ESS criteria met ○ MCSE or ESS criterion not met

Function names refer to the package above each panel unless another library is named. Added here identifies our code. PFJAX OT differentiation failed; the plotted correction was added here.

Purple methods in PFJAX, PyDPF, TFP and Dynamax: alpha 1, 425 Adam updates without IF2 initialization. PyPOMP / Cuthbert IFAD alpha 1 uses 60 IF2 + 425 Adam updates.

Boxplots and medians use all finite estimates; only points outside Q1 - 1.5 IQR or Q3 + 1.5 IQR are hidden. Bars show +/- 1.96 MCSE. Counts beneath each method give shown/finite estimates.

69 of 961 finite estimates are hidden; 892 are shown. Nine fits failed, seven without a finite score; all four estimates near -24,868 are hidden. Full results remain in overview_v1.